Billy Jhoe F. Gacias **Activity 4 - Base Conversion Laboratory** 09/24/2026

BSCS 1-1

Binary

Division	Quotient	Remainder
918 ÷ 2	459	0
459 ÷ 2	229	1
229 ÷ 2	114	1
114 ÷ 2	57	0
57 ÷ 2	28	1
28 ÷ 2	14	0
14 ÷ 2	7	0
7 ÷ 2	3	1
3 ÷ 2	1	1
1 ÷2	0	1

Read bottom to top: 1110010110_2

Octal

Division	Quotient	Remainder
918 ÷ 8	114	6
144 ÷ 8	14	2
14 ÷ 8	1	6
1 ÷ 8	0	1

Read bottom to top: 1626_8

Division	Quotient	Remainder
918 ÷ 16	57	6
57 ÷ 16	14	9
3 ÷ 16	0	3

Read bottom to top: 396_{16}

To Octal

3-bit group (left to right)	001	110	010	110
Octal digit	1	6	2	6

1626_8

4-bit group (left to right)	0011	1001	0110
Hexadecimal digit	3	9	6

396_{16}

Binary

Position Value	2^9 (512)	2^8 (256)	2^7 (128)	2^6 (64)	2^5 (32)	2^4 (16)	2^3 (8)	2^2 (4)	2^1 (2)	2^0 (1)
Binary Bit	1	1	1	0	0	1	0	1	1	0

512+256+128+0+0+16+0+4+2+0 = 918

Position Value	8^3 (512)	8^2 (64)	8^1 (8)	8^0 (1)
Octal Digit	1	6	2	6

(1 * 512) + (6 * 64) + (2 * 8) + (6 * 1) = 512 + 384 + 16 + 6 = 918

Position Value	16^2 (512)	16^1 (16)	16^0 (1)
Hex Digit	3	9	6

(3 * 256) + (9 * 16) + (6 * 1) = 768 + 144 + 6 = 918